

Plant High Molecular Weight (HMW) DNA Manual Extraction Protocol:

Updated September 2024

Reagents:

- **Sorbitol wash** (from Jones and Schwessinger, 2020):
 - 0.35 M D-Sorbitol
 - 1% (w/v) PVP-40
 - 100 mM Tris-HCl (pH8)
 - 5 mM EDTA pH8 DTT (1,4-Dithiothreitol).
- **Lysis Buffer** (Schalamun et al., 2018):
 - 1% w/v PVP-40
 - 1% Sodium metabisulfite
 - 0.5 M Sodium Chloride
 - 0.1 M Tris-HCl pH 8.0
 - 0.05 mM EDTA, pH 8.0
 - 1.5% (w/v) Sodium dodecyl sulfate (SDS).
- 1% β -Mercaptoethanol (BME)
- 5M Potassium Acetate
- 100% Chloroform
- 20 mg/mL RNase A
- 100% Isopropanol
- 70% Ethanol
- Nuclease free dH₂O

Equipment:

- Centrifuge
- Water bath
- Hulamixer
- 50 mL Nalgene centrifuge tubes

Consumables:

- 1.5 mL DNA LoBind Eppendorf tubes
- 50 mL Falcon tubes
- P1000 wide bore tips
- P200 wide bore tips
- P1000 tips
- P200 tips
- P10 tips

HMW Genomic DNA Extraction Protocol**Sampling:**

1. Cut plant leaf material into a pre-cooled mortar filled with liquid nitrogen.
2. Pulverize 1g of plant tissue (per 50 ml Nalgene tube) in liquid nitrogen. (Grind for at least 25 min until leaf material is a smooth, homogenous fine powder). Transfer 1 g of tissue to a pre-cooled 50mL Nalgene extraction tube.

Sorbitol Wash:

3. Prepare the sorbitol wash solution (D-Sorbitol 0.35 M, PVP-40 000 1% (w/v), Tris-HCl pH8 100 mM, EDTA pH8 5 mM) as per Jones and Schwessinger, (2020) beforehand.
4. Prepare ~150 mL sorbitol wash per 1g tissue prepared for extraction. Add 3x small scoops of DTT to 50 mL of the sorbitol wash in a 50 mL Falcon tube solution just before use.

5. Resuspend 1 g of ground tissue in 25 mL sorbitol wash solution in the 50 mL Nalgene tube. Make sure the tissue is homogenised in the sorbitol solution by shaking vigorously. Make sure all the tissue is in solution without any lumps.
6. Centrifuge at 5000g, room temperature for 5 min (time can be adjusted if the working with a plant that has a loose pellet to prevent the loss of tissue).
7. Pour off the supernatant and repeat step 5 until supernatant is clear (minimum of 6X).
8. During the final sorbitol wash, resuspend the tissue in exactly 15 mL of sorbitol wash solution. Once tissue is homogenous in solution, transfer 1.5 mL of homogenous tissue + sorbitol solution to a 1.5 mL DNA LoBind Eppendorf tube (total of 10X 1.5 mL tubes per 1 g tissue) using a P1000 pipette and wide bore pipette tip.
9. Centrifuge at 5000g, room temperature for 5 min.
10. Pour off the supernatant.

Cell Lysis:

11. Pre-mix Lysis buffer and RNase A (20 mg/mL) for 10x 1.5mL Eppendorf tubes: 10 ml Lysis buffer and 20 μ L RNase A (20 mg/ml) in a 50 mL Falcon tube.
12. Add 800 μ L of the pre-mixed Lysis buffer to each 1.5 mL Eppendorf tube. Mix gently using a P1000 pipette and a wide bore tip by pulsating the pipette 2X and gently flicking the tube until pellet is dissolved and solution is homogenous.
13. Incubate at 37°C for 30 minutes (possible to use lukewarm water).
14. Add 8 μ L of BME to each tube and invert tubes 3X.
15. Incubate at 65°C in a water bath for 10 minutes. Mix every 2 minutes through gentle inversion.
16. Remove samples from water bath and gently invert tubes 3X to get everything into solution.
17. Place samples on ice for 4 minutes to cool down.
18. Add 150 μ L Potassium acetate to each tube, invert tubes 3X and place back on ice for 20 minutes.

DNA Extraction:

19. Add 500 μ L of ice-cold chloroform to samples.
20. Shake well to get everything homogenous and place on Hulamixer with 360° rotation at room temperature for 10 min. Centrifuge for 10 min @1300g, room temperature.
21. Transfer the aqueous phase using a P200 pipette and wide bore pipette tips and pipetting slowly to a new 1.5 mL DNA LoBind Eppendorf tube.
22. If aqueous phase appears very murky after centrifugation step, add 150 μ L Potassium acetate to each tube and then repeat steps 19 to 21.
23. Add an equal volume (700 μ L) of isopropyl alcohol previously cooled to -20 °C. Mix gently by slowly inverting tubes 5X. Place the samples at -20 °C. for 1h.
24. Centrifuge for 5 min at 4000 g, room temperature (place Eppendorf tubes with the back of tube facing to the back of the rotor to help locate pellet). Discard the supernatant carefully. Be careful not to lose your pellet; some may be transparent and stuck to the site of the tube.
25. Resuspend the pellet in 1 mL of 70% ethanol previously cooled to -20°C. Mix gently by flicking and inverting tubes 5X. Make sure the pellet dislodges from the bottom.
26. Centrifuge for 5 min at 4000 g, room temperature (place Eppendorf tubes with back of tube facing to the back of the rotor to help locate pellet). Discard the supernatant gently.
27. Repeat steps 25 and 26.
28. After a quick spin down, immediately take off excess ethanol using a pipette without touching the pellet. Allow to dry for max 30 seconds.
29. Resuspend the pellet in 13 -23 μ L (depending on the size of the pellet) of nuclease free dH₂O.
30. Proceed to Quality Control steps. Do not freeze DNA, store at 4 °C.

References:

Ashley Jones, Benjamin Schwessinger 2020. Sorbitol washing complex homogenate for improved DNA extractions. **protocols.io** <https://dx.doi.org/10.17504/protocols.io.beuvjew6>

Schalamun M, Nagar R, Kaine D, et al. Harnessing the MinION: An example of how to establish long-read sequencing in a laboratory using challenging plant tissue from *Eucalyptus pauciflora*. *Mol Ecol Resour.* 2019; 19: 77–89. <https://doi.org/10.1111/1755-0998.12938>